

7. (3.25 points)

Given the following information for a company as of December 31, 2018:

Cumulative Case Incurred Losses (\$000,000) as of (months)						
Accident Year	12	24	36	48	60	72
2013	3,384	4,282	4,813	5,408	5,394	5,362
2014	3,499	4,438	5,646	5,532	5,512	
2015	3,483	5,566	5,539	5,398		
2016	6,502	6,197	6,222			
2017	6,656	6,658				
2018	6,595					

Case Incurred Loss Development Factors as of (months)					
Accident Year	12-24	24-36	36-48	48-60	60-72
2013	1.265	1.124	1.124	0.997	0.994
2014	1.268	1.272	0.980	0.996	
2015	1.598	0.995	0.975		
2016	0.953	1.004			
2017	1.000				

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7. (continued)

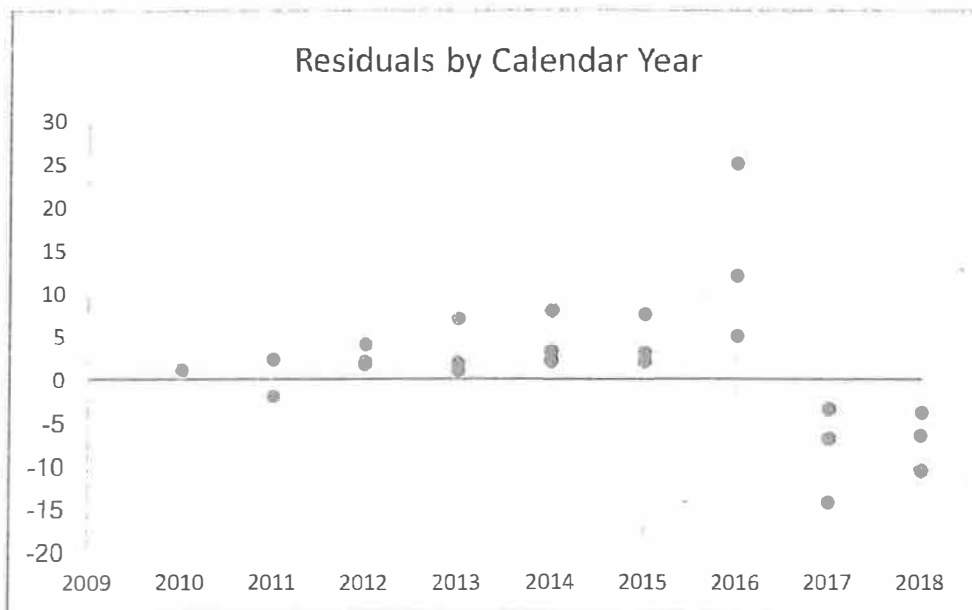
a. (1.75 points)

Test the null-hypothesis that there are no calendar year influences in the data using Mack's methodology assuming an error probability of 5% where:

- $E[Z] = 3.25$
- $\text{Var}(Z) = 1.06$
- The 97.5th percentile of the Standard Normal distribution is 1.96.

b. (1.5 points)

The graph below shows the weighted residuals of case incurred losses for maturities of 24, 36, and 48 months.



Discuss whether the results above could be explained by each of the following:

- Sustained increase in calendar year trend rate in 2016.
- Increase in case reserve adequacy in 2016.
- Acceleration of loss payment in 2016.

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[illegible]

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Sample 3

	12-24	24-36	36-48	48-60	60-72
2013	*	L	L	L	*
2014	L	L	*	S	
2015	L	S	S		
2016	S	S			
2017	S				

$$Z^* = 0 + 0 + 0 + 1 + 0 = 1$$

$$C.I.: (3.25 - 1.96\sqrt{1.06}, 3.25 + 1.96\sqrt{1.06}) \\ = (1.232, 5.268)$$

$Z^* = 1$ is not in the 95% C.I., we can therefore reject the null hypothesis that there are no calendar year influences.

Sample 4

AY	12-24	24-36	36-48	48-60	60-72	
2013	*	H	H	H	*	* = median in column
2014	H	H	*	L		H = above col median
2015	H	L	L			L = below col median
2016	L	L				
2017	L					$Z_i = \min\{\#H, \#L\}$ in i^{th} diagonal

$$\begin{aligned} Z_1 &= 0 \\ Z_2 &= 0 \\ Z_3 &= 0 \\ Z_4 &= 1 \\ Z_5 &= 0 \end{aligned} \quad \begin{aligned} Z &= \sum Z_i = 1 \\ Z \pm 1.96sd(Z) &= 1 \pm 1.96\sqrt{1.06} \\ &= -1.018, 3.0179 \end{aligned}$$

95% CI excludes $E(Z)$, so reject null! Conclude there are significant CY effects.

Sample 5

Triangle of rank, with median circled

AY	12-24	24-36	36-48	48-60	60-72
2013	③	3	3	2	①
2014	4	4	②	1	
2015	5	1	1		
2016	1	2			
2017	2				

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Triangle of smaller vs larger

CY	AY	12-24	24-36	36-48	48-60	60-72
1	2013	*	L	L	L	*
2	2014	L	L	*	S	
3	2015	L	S	S		
4	2016	S	S			
5	2017	S				

CY	nb S	nb L	Z
1	0	0	0
2	0	2	$\min(0,2) = 0$
3	0	3	$\min(0,3) = 0$
4	2	1	$\min(2,1) = 1$
5	4	0	$\min(4,0) = 0$
			$0 + 0 + 0 + 1 + 0 = 1$

We have $Z \text{ score} = \frac{Z - E[Z]}{\sqrt{\text{Var}[Z]}} = \frac{1 - 3.25}{\sqrt{1.06}} = -2.1854$

Since $|Z\text{-score}| = 2.1854 > 1.96$, we reject H_0 .

There are no calendar year effects.

Part b: 1.50 points

Sample Responses for [i]

- No. This would cause losses in CY2017 & 2018 to be higher than expected, resulting in positive/higher residuals, not negative like in graph.
- No, a sustained calendar year trend would show consistent under prediction and not the reversal we see in 2017 and 2018.
- A sustained trend increase would imply 2017 & 2018 actuals are also greater than expected, so this does not explain the data.
- If we experienced a CY trend increase in 2016 and beyond then we would expect our actuals to be above expected so we should expect positive residuals in 2017 & 2018. Therefore this does not explain the results we see.

Sample Responses for [ii]

- Yes. If case reserves are larger (more adequate), Mack model will overestimate development, resulting in negative residuals in CY2017 and 2018.
- Yes. A sudden increase in case reserves would initially show underprediction. If 2017 and 2018 CR return to normal, the large 2016 losses would increase LDF which would overpredict 17 and 18.
- Strengthening of case reserves in 2016 would cause positive residuals as shown in the graph. It would also lead to less future incurred, so 2017 and 2018 would have negative residuals. Thus this occurrence would explain the results in the graph.
- An increase in case reserve adequacy in 2016 would result in higher actual incurreds in CY 2016 and therefore higher positive residuals in 2016, which we see in the graph. If this increased case reserve adequacy did not persist in 2017 and 2018 when we have negative residuals, then this could explain the result in we see.

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- Yes, residuals are high in 2016 showing the reserve increase, then low in later years since reserves will develop less going forward.

Sample Responses for [iii]

- No. This impacts paid, rather than reported claims, and this data is for reported losses.
- No. The triangle is based on incurred data. Assuming case reserves are adequate, an acceleration at payments will not affect incurred development.
- If claims are adequately reserved, acceleration of loss payments would not have an effect on incurred losses and would thus not explain the results.
- An acceleration in loss payment wouldn't have any effect on these residuals because these are for incurred losses. As long as the increase in paid is offset by reduction to case reserves then we wouldn't necessarily see any CY impact to the residuals. Therefore does not explain.
- No. If losses are paid faster, but the case incurred amount does not change, residuals here shouldn't be impacted.

EXAMINER'S REPORT

Candidates were expected to understand and demonstrate how to perform Mack's methodology of testing for calendar year influences in data, and to analyze a graph of case incurred residuals for the type of change that could create a certain residual pattern.

Part a

Candidates were expected to convert the triangle of case incurred loss development factors into a triangle of large/small, hi/low, or similar indicators consistent with Mack's methodology, then calculate the test statistic Z as the sum of the minimum of the count of large and small indicators on each diagonal. Candidates were then expected to calculate the confidence interval for the test statistic, compare the test statistic to the confidence interval, and reject the null hypothesis since the test statistic falls outside the confidence interval. Since the null hypothesis was explicitly provided in the question, candidates were not required to restate the null hypothesis in their conclusion for full credit.

Some candidates failed to notice that $E[Z]$ and $\text{Var}[Z]$ were provided in the question and calculated them directly. While candidates did not lose credit for this so long as the values were calculated correctly, they may have lost valuable time.

Common mistakes included:

- Incorrectly assigning large and small indicators.
- Applying large and small indicators to the case incurred triangle instead of the case incurred development factor triangle.
- Using a calculation of Z other than the minimum of the count of S and L indicators by diagonal, such as the maximum.
- Not taking the square root of the variance when calculating the range.
- Providing a conclusion that was made without comparison of the test statistic to the confidence interval.
- Misstating the null hypothesis in the conclusion. Examples include stating that the null hypothesis is that there are calendar year influences, that calendar years are independent, or that columns of development factors are uncorrelated.

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- Performing a different test, such as Mack's test for correlation of columns of development factors.

Part b

Candidates were expected to understand the effects of the following and to make relevant observations from the graph to support their conclusions:

- A sustained increase in calendar year trend.
- An increase in case reserve adequacy.
- An acceleration of loss payments on case incurred loss development and residuals.

When discussing a sustained increase in calendar year trend or an increase in case reserve adequacy, candidates needed to discuss calendar years 2017 and 2018 in order to receive full credit.

When discussing the impact of an acceleration in loss payments, some candidates stated that residuals are, or should be, positive or negative. This was not considered a sufficient response because an increase in payments should be offset by a reduction in case reserves, resulting in no impact on case incurred all other factors being constant.

Common mistakes included:

- Not supporting their conclusion by relating it to the graph
- Not addressing calendar years 2017 and 2018
- Not understanding that calendar years are incremental
- Not understanding that an increase in case reserve adequacy would cause residuals for 2017 and 2018 to decrease, following the increase in 2016
- Trying to draw conclusions about payment patterns from case incurred residuals
- Misinterpreting case incurred as case reserves
- Misinterpreting the x-axis as Accident Year instead of Calendar Year